Revisiting finite deformation poromechanics: Deriving a nonlinear Biot coefficient from first principles

John T. Foster*

The Hildebrand Department of Petroleum & Geosystems Engineering
Department of Aerospace Engineering & Engineering Mechanics
The Oden Institute for Computational Engineering & Science
The University of Texas at Austin

Xiao Xu

The Oden Institute for Computational Engineering & Science
The University of Texas at Austin

June 22, 2026

Abstract

This paper presents a novel framework that integrates mixture theory with finite deformation poromechanics to derive a nonlinear Biot coefficient from first principles, eliminating the need for constitutive assumptions prevalent in prior formulations. By applying the extended Hamilton's principle to a binary mixture of solid and fluid phases, we fully account for finite deformations and employ a Legendre transformation to shift the internal energy dependence from specific volume to pressure. This approach naturally yields a new nonlinear form of the Biot coefficient that is shown to reduce to the classical Biot coefficient under small deformations. Our method provides a clear physical interpretation of the Biot coefficient as a measure of the solid's specific volume change with respect to skeleton deformation at constant pore pressure. This work bridges mixture theory and finite deformation poromechanics, offering a fundamental derivation that enhances the understanding and applicability of poromechanical models in large deformation scenarios.

1 Introduction

Poromechanics provides an essential framework for modeling fluid-saturated porous media, with applications spanning geomaterials, biomaterials, and engineered systems. The foundational work of Biot [3] established this field under small deformation assumptions, offering a robust basis for coupling solid and fluid phases in a continuum mixture. However, many real-world scenarios—such as landslides, tissue mechanics, or advanced porous materials—involve large deformations that exceed the limits of Biot's classical theory. Finite deformation poromechanics addresses these challenges, but introduces complexities in defining key parameters, often relying on constitutive assumptions to bridge solid-fluid interactions.

*Electronic address: john.foster@utexas.edu

Significant advancements, such as those by Coussy [5] and Borja [4], have extended poromechanics into the finite deformation regime by integrating mixture theory with thermodynamic or micromechanical principles. Borja's work, in particular, offers a sophisticated approach, deriving a nonlinear Biot coefficient linked to the solid skeleton's bulk modulus. His formulation, grounded in thermodynamics provides a mathematically consistent extension of effective stress concepts including to partially saturated media. However, it depends on constitutive assumptions–such as an assumed, but undefined, constitutive relationship between pressure, density, and volume fraction–that can obscure physical interpretation and complicate experimental validation. While Borja's method is rigorous and valuable, its reliance on such assumptions highlights an opportunity for alternative perspectives that minimize imposed relationships.

In this paper, we introduce a novel framework that unifies mixture theory with finite deformation poromechanics, deriving a nonlinear Biot coefficient,

$$B = 1 - \frac{1}{v_0^s} \frac{\partial \bar{v}^s}{\partial J^s}\bigg|_p ,$$

directly from first principles. Here, $v_0^s = 1/\rho_0^s$ is the reference specific volume (inverse density) of the solid, $\bar{v}^s = 1/\bar{\rho}^s$ is the current *local* specific volume, J^s is the solid skeleton's Jacobian determinant, and p is the pore pressure – held fixed. Building on the variational method of Bedford and Drumheller [2], we apply the extended Hamilton's principle to a binary mixture of solid and fluid phases, fully accounting for finite deformations. Unlike traditional approaches, our method avoids constitutive assumptions about the Biot coefficient's form, such as those based on constitutive introduction into the entropy inequalities (i.e. during Coleman-Noll procedures) or effective stress definitions.

A cornerstone of our approach is the use of a Legendre transformation within the variational formulation, shifting the internal energy dependence from specific volume to pressure. This transformation naturally yields the Biot coefficient as a measure of the solid's specific volume change with respect to skeleton deformation at constant pore pressure. Compared to Borja's framework, where the coefficient emerges from a complex expression tied to assumed constitutive functions, our derivation offers distinct advantages. First, it provides a clear physical interpretation—suggesting straightforward experimental validation, such as tracking density changes via microCT—without requiring abstract constitutive relationships. Second, it extends seamlessly to finite deformations, capturing nonlinear effects through J^s, while aligning with the classical form $1 - K/K^s$ in the small-deformation limit, consistent with Borja's findings and the earlier work of Skempton [8] and Nur and Byerlee [7]. Our approach complements these previous works by offering a more fundamental derivation that minimizes assumptions, enhances experimental accessibility, and clarifies the coefficient's role as a material property evaluated at constant pressure—a distinction less explicit in Borja's formulation. This framework aligns with variational principles in other finite deformation studies, such as Armero [1], yet stands apart by deriving the Biot coefficient organically, without presupposing its existence. To our knowledge, this work is the first that links mixture theory or the thermodynamic description of continuum mixtures with finite deformation poromechanics, elucidating the Biot coefficient without constitutive assumptions.

The paper is structured as follows: Section 2 establishes continuum mechanics concepts within mixture theory; Section 3 derives momentum equations using the extended Hamilton's principle following closely the work of Bedford and Drumheller [2]; Section 4 validates our equations against Biot's theory under small deformations and interprets the effective stress; and we conclude by exploring the physical significance of our nonlinear Biot coefficient. In this work, we focus primarily on deriving the momentum balance equations, as the mass balance remains consistent with estab-

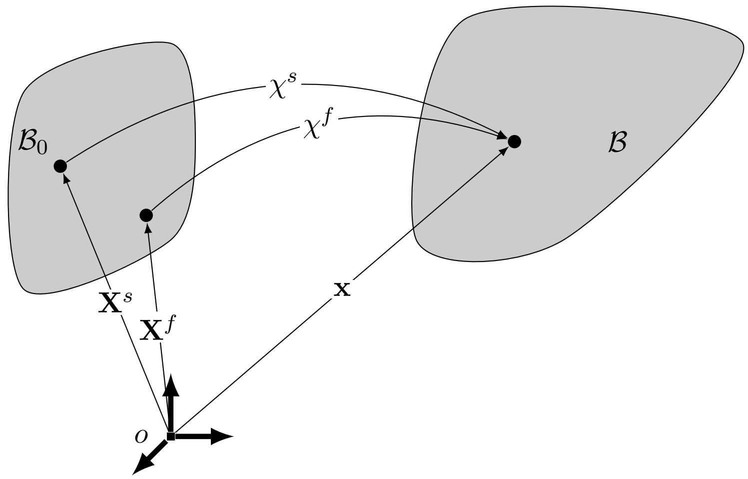

Figure 1: Schematic description of kinematics

lished formulations in the literature. This allows us to concentrate on the novel aspects of our approach.

2 Basic Concepts of Continuum Mechanics in Mixture Theory

Consider a continuum body which is an immiscible mixture of two constituents (solid and fluid). According to mixture theory, these two constituents simultaneously occupy the same material point identified by a position vector $\mathbf{x}$ in the current (deformed) configuration of a body $\mathcal{B}$. In general, the two constituents, solid and fluid, can move independent of one another from their own position vectors $\mathbf{X}^s$ and $\mathbf{X}^f$ in the reference (undeformed) configuration of the body $\mathcal{B}_0$. Let χ^s and χ^f denote the mapping between the reference and current configuration for the solid and fluid respectively and assume that χ^s and χ^f are sufficiently smooth and one-to-one mappings. Based on these definitions, we can write

$$\mathbf{x} = \chi^\alpha(\mathbf{X}^\alpha, t), \qquad \alpha = s, f$$

where the superscripts s and f represent the solid and fluid respectively. From continuum mechanics, the deformation gradient $\mathbf{F}^\alpha$ for each constituent is then defined as

$$\mathbf{F}^\alpha = \frac{\partial \chi^\alpha}{\partial \mathbf{X}^\alpha}.$$

Following arguments from vector calculus, the infinitesimal volume occupied by the constituent α in the reference configuration $\mathrm{d}V_0^\alpha$ is related to the infinitesimal volume occupied by all constituents in the current configuration $\mathrm{d}v$ through the Jacobian determinant J^α:

$$\mathrm{d}v = J^\alpha \, \mathrm{d}V_0^\alpha, \tag{1}$$

where the Jacobian determinant J^α is given by

$$J^\alpha = \det(\mathbf{F}^\alpha).$$

Assume that both constituents occupy an identical volume $\mathcal{B}_0$ with the same surface $\partial\mathcal{B}_0$ at time t_1 and another identical volume $\mathcal{B}$ with the same surface $\partial\mathcal{B}$ at time t_2, or, in other words, they do not separate at the boundaries but are allowed to move relative to one another internally, over a time interval $[t_1, t_2]$. Then, mass conservation of the body can be expressed as

$$\int_{\mathcal{B}_0} \rho_0^\alpha(\mathbf{X}^\alpha)\, \mathrm{d}V_0^\alpha = \int_{\mathcal{B}} \rho^\alpha(\mathbf{x}, t)\mathrm{d}v, \tag{2}$$

where ρ^α is the partial density defined as the mass of α constituent divided by the total volume of all constituents in the current configuration. The subscript 0 is used to indicate evaluation in the reference configuration. Substitute equation (1) into (2), we have

$$\int_{\mathcal{B}_0} \rho_0^\alpha\, \mathrm{d}V_0^\alpha = \int_{\mathcal{B}_0} \rho^\alpha J^\alpha\, \mathrm{d}V_0^\alpha. \tag{3}$$

Equation (3) must hold for any arbitrary volume, therefore we can conclude

$$J^\alpha = \frac{\rho_0^\alpha}{\rho^\alpha}.$$

It is useful to define a local density for each constituent. The local densities $\bar{\rho}^\alpha$, which are defined as the mass of the α constituent divided by the volume of the α constituent in the current configuration, are related to the partial densities ρ^α by

$$\rho^\alpha = \phi^\alpha(\mathbf{x}, t)\bar{\rho}^\alpha(\mathbf{x}, t),$$

where ϕ^α is the volume fraction of the α constituent which is defined as the volume of the α constituent over the total volume at any given material point again in the current configuration. Note that the definition of volume fraction leads to an equation called the *volume fraction constraint*

$$\sum_{\alpha=s,f} \phi^\alpha = 1. \tag{4}$$

With these definitions of partial densities and volume fractions, our mass conservation equation can be further written as

$$J^\alpha = \frac{\phi_0^\alpha \bar{\rho}_0^\alpha}{\phi^\alpha \bar{\rho}^\alpha}. \tag{5}$$

3 Conservation of Momentum using Extended Hamilton's Principle

The extended Hamilton's principle states that among admissible motions, the actual motion of a body is such that

$$\int_{t_1}^{t_2} \delta(T - V) + \delta W + \sum_k \delta C_k \mathrm{d}t = 0, \tag{6}$$

where δ is the variation operator from the calculus of variations, T is the kinetic energy of the body, V is the potential energy, δW is the virtual work done by external forces on the body, C_k are constraints equations restricting the motion; and t_1, t_2 are the fixed times in which the configuration of the body is assumed to be prescribed.

We first focus on the constraints imposed on the porous media. As discussed in the previous section, there are two constraints: one is the mass conservation equations (5), the other is the volume fraction constraint (4), denoted by C_1 and C_2 respectively

$$C_1 = \sum_{\alpha=s,f} \int_{\mathcal{B}_0} \lambda^\alpha(\mathbf{X}^\alpha, t) \left(J^\alpha - \frac{\phi_0^\alpha \bar{\rho}_0^\alpha}{\phi^\alpha \bar{\rho}^\alpha} \right) \mathrm{d}V_0^\alpha, \tag{7}$$

$$C_2 = \int_{\mathcal{B}} p(\mathbf{x}, t) \left(\sum_{\alpha=s,f} \phi^\alpha - 1 \right) \mathrm{d}v, \tag{8}$$

where λ^α and p are Lagrange multipliers. $\lambda^\alpha(\mathbf{X})$ is defined on the reference configuration because the constraint (mass conservation) is expressed in the reference configuration while $p(\mathbf{x})$ is defined in the current configuration given that the volume fraction constraint must be satisfied at every material point in the current configuration.

Taking the first variation of equation (7) using $\mathbf{x}$, ϕ^α, and $\bar{\rho}^\alpha$ as independent fields gives[1]

$$\begin{aligned}
\delta C_1 &= \sum_{\alpha=s,f} \int_{\mathcal{B}_0} \lambda^\alpha \, \delta J^\alpha - \lambda^\alpha \, \delta \left(\frac{\phi_0^\alpha \bar{\rho}_0^\alpha}{\phi^\alpha \bar{\rho}^\alpha} \right) \mathrm{d}V_0^\alpha, \\
&= \sum_{\alpha=s,f} \int_{\mathcal{B}_0} \lambda^\alpha J^\alpha (F_{ji}^\alpha)^{-1} \frac{\partial \delta x_i^\alpha}{\partial X_j^\alpha} + \lambda^\alpha J^\alpha \left(\frac{\delta \phi^\alpha}{\phi^\alpha} + \frac{\delta \bar{\rho}^\alpha}{\bar{\rho}^\alpha} \right) \mathrm{d}V_0, \\
&= \sum_{\alpha=s,f} \int_{\mathcal{B}} \lambda^\alpha \frac{\partial \delta x_i^\alpha}{\partial x_j} + \lambda^\alpha \left(\frac{\delta \phi^\alpha}{\phi^\alpha} + \frac{\delta \bar{\rho}^\alpha}{\bar{\rho}^\alpha} \right) \mathrm{d}v, \\
&= \sum_{\alpha=s,f} \int_{\partial\mathcal{B}} \lambda^\alpha \hat{n}_i \delta x_i^\alpha \mathrm{d}s - \int_{\mathcal{B}} \frac{\partial \lambda^\alpha}{\partial x_j} \delta x_i^\alpha + \lambda^\alpha \left(\frac{\delta \phi^\alpha}{\phi^\alpha} + \frac{\delta \bar{\rho}^\alpha}{\bar{\rho}^\alpha} \right) \mathrm{d}v, \\
&= \sum_{\alpha=s,f} \int_{\partial\mathcal{B}} \lambda^\alpha \hat{\mathbf{n}} \cdot \delta \mathbf{x}^\alpha \mathrm{d}s - \int_{\mathcal{B}} \nabla_{\mathbf{x}} \lambda^\alpha \cdot \delta \mathbf{x}^\alpha + \lambda^\alpha \left(\frac{\delta \phi^\alpha}{\phi^\alpha} + \frac{\delta \bar{\rho}^\alpha}{\bar{\rho}^\alpha} \right) \mathrm{d}v,
\end{aligned} \tag{9}$$

where integration-by-parts has been used along with the identities

$$\begin{aligned}
\delta J^\alpha =& \frac{\partial J^\alpha}{\partial F_{ij}^\alpha} \delta F_{ij}^\alpha, \\
=& \frac{\partial J^\alpha}{\partial F_{ij}^\alpha} \frac{\partial \delta x_i^\alpha}{\partial X_j^\alpha}, \\
=& J^\alpha (F_{ji}^\alpha)^{-1} \frac{\partial \delta x_i^\alpha}{\partial X_j^\alpha},
\end{aligned} \tag{10}$$

and

$$\begin{aligned}
\delta \left(\frac{\phi_0^\alpha \bar{\rho}_0^\alpha}{\phi^\alpha \bar{\rho}^\alpha} \right) =& \delta \left((\phi_0^\alpha \bar{\rho}_0^\alpha)(\phi^\alpha \bar{\rho}^\alpha)^{-1} \right), \\
=& (\phi_0^\alpha \bar{\rho}_0^\alpha) \delta (\phi^\alpha \bar{\rho}^\alpha)^{-1}, \\
=& - J^\alpha (\phi^\alpha \bar{\rho}^\alpha)^{-1} (\delta \phi^\alpha \bar{\rho}^\alpha + \phi^\alpha \delta \bar{\rho}^\alpha), \\
=& - J^\alpha \left(\frac{\delta \phi^\alpha}{\phi^\alpha} + \frac{\delta \bar{\rho}^\alpha}{\bar{\rho}^\alpha} \right).
\end{aligned} \tag{11}$$

[1] We will readily switch between indicial and vector notation when needed for utility in deriving equations or for clarity.

Equation (9) is the variation of the first constraint. For the second constraint (8), it is non-trivial to compute variations of $\phi^\alpha(\mathbf{x})$ in the current configuration because ϕ^α is a function of $\mathbf{x}$ which is also varied. To proceed with this variation we will use the Gâteaux derivative approach [6] after a pull-back to the reference configuration. To illustrate this, we first work on the variation of the following equation

$$\sum_\alpha \phi^\alpha(\mathbf{x}) = 1. \tag{12}$$

Suppose that there are a small variation $\epsilon\delta\phi^\alpha$ on the volume fraction ϕ^α and also a small variation $\epsilon\delta\mathbf{x}^\alpha$ on the current position $\mathbf{x}^\alpha$. Then define

$$\phi^{*\alpha} = \phi^\alpha + \epsilon\delta\phi^\alpha,$$

and

$$\begin{aligned}\mathbf{x}^{*\alpha} =& \chi^\alpha(\mathbf{X}^\alpha) + \epsilon\delta\mathbf{x}^\alpha,\\ =& \boldsymbol{\Upsilon}(\mathbf{X}^\alpha, \epsilon),\end{aligned}$$

Also define the inverse

$$\mathbf{X}^\alpha = \boldsymbol{\Upsilon}^{-1}(\mathbf{x}^{*\alpha}, \epsilon). \tag{13}$$

With these definitions, the first variation of (12) can be written as the Gâteaux derivative holding $\mathbf{x}^{*\alpha}$ fixed.

$$\begin{aligned}0 =& \sum_\alpha \frac{\mathrm{d}}{\mathrm{d}\epsilon}\left[\phi^\alpha(\mathbf{X}^\alpha) + \epsilon\delta\phi^\alpha\right]|_{\epsilon=0},\\ =& \sum_\alpha \frac{\mathrm{d}}{\mathrm{d}\epsilon}[\phi^\alpha(\boldsymbol{\Upsilon}^{-1}(\mathbf{x}^{*\alpha}, \epsilon)) + \epsilon\delta\phi^\alpha]|_{\epsilon=0},\\ =& \sum_\alpha \left(\left[\frac{\partial\phi^\alpha}{\partial\boldsymbol{\Upsilon}^{-1}}\frac{\partial\boldsymbol{\Upsilon}^{-1}}{\partial\epsilon}\right]\bigg|_{\epsilon=0} + \delta\phi^\alpha\right).\end{aligned} \tag{14}$$

In order to evaluate the term $\frac{\partial\boldsymbol{\Upsilon}^{-1}}{\partial\epsilon}$, compute the differential holding $\mathbf{X}^\alpha$ fixed on both sides of equation (13):

$$\mathbf{0} = \frac{\partial\boldsymbol{\Upsilon}^{-1}}{\partial\mathbf{x}^{*\alpha}}\mathrm{d}\mathbf{x}^{*\alpha} + \frac{\partial\boldsymbol{\Upsilon}^{-1}}{\partial\epsilon}\mathrm{d}\epsilon,$$

and solve

$$\begin{aligned}\frac{\partial\boldsymbol{\Upsilon}^{-1}}{\partial\epsilon} =& -\frac{\partial\boldsymbol{\Upsilon}^{-1}}{\partial\mathbf{x}^{*\alpha}}\frac{\partial\mathbf{x}^{*\alpha}}{\partial\epsilon}\bigg|_{\mathbf{X}^\alpha},\\ =& \frac{\partial\boldsymbol{\Upsilon}^{-1}}{\partial\mathbf{x}^{*\alpha}}\delta\mathbf{x}^\alpha.\end{aligned} \tag{15}$$

Substituting (15) into (14) gives

$$\begin{aligned}=& \sum_\alpha \left(\left[-\frac{\partial\phi^\alpha}{\partial\boldsymbol{\Upsilon}^{-1}}\frac{\partial\boldsymbol{\Upsilon}^{-1}}{\partial\mathbf{x}^{*\alpha}}\cdot\delta\mathbf{x}^\alpha\right]\bigg|_{\epsilon=0} + \delta\phi^\alpha\right),\\ =& \sum_\alpha \left(\left[-\frac{\partial\phi^\alpha}{\partial\mathbf{x}^{*\alpha}}\cdot\delta\mathbf{x}^\alpha\right]\bigg|_{\epsilon=0} + \delta\phi^\alpha\right),\\ =& \sum_\alpha \left(-\frac{\partial\phi^\alpha}{\partial\mathbf{x}}\cdot\delta\mathbf{x}^\alpha + \delta\phi^\alpha\right),\\ =& \sum_\alpha \left(\nabla_{\mathbf{x}}\phi^\alpha\cdot\delta\mathbf{x}^\alpha - \delta\phi^\alpha\right).\end{aligned} \tag{16}$$

Substituting this result (16) into the volume fraction constraint (8) gives us the result for variation of the second constraint:

$$\delta C_2 = \sum_{\alpha=s,f} \int_{\mathcal{B}} p \left(\nabla_{\mathbf{x}} \phi^{\alpha} \cdot \delta \mathbf{x} - \delta \phi^{\alpha}\right) \mathrm{d}v. \tag{17}$$

Next, we evaluate the variation of the rest of terms in (6). The total kinetic energy of the mixture can be written as

$$T = \frac{1}{2} \int_{\mathcal{B}} \rho^s \mathbf{v}^s \cdot \mathbf{v}^s + \rho^f \mathbf{v}^f \cdot \mathbf{v}^f \mathrm{d}v,$$

where $\mathbf{v}^s$ and $\mathbf{v}^f$ are the velocities of solid and fluid respectively:

$$\mathbf{v}^{\alpha} = \frac{\partial \chi^{\alpha}(\mathbf{X}^{\alpha}, t)}{\partial t}, \qquad \alpha = s, f$$

Taking the first variation of the kinetic energy during the application of Hamilton's principle and integrating in time give us

$$\begin{aligned}
\int_{t_1}^{t_2} \delta T \mathrm{d}t &= \int_{t_1}^{t_2} \frac{\partial T}{\partial \mathbf{v}^s} \cdot \delta \mathbf{v}^s + \frac{\partial T}{\partial \mathbf{v}^f} \cdot \delta \mathbf{v}^f \mathrm{d}t, \\
&= \left[\frac{\partial T}{\partial \mathbf{v}^s} \cdot \delta \mathbf{x}^s + \frac{\partial T}{\partial \mathbf{v}^f} \cdot \delta \mathbf{x}^f \right] \Bigg|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial T}{\partial \mathbf{v}^s} \right) \cdot \delta \mathbf{x}^s + \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial T}{\partial \mathbf{v}^f} \right) \cdot \delta \mathbf{x}^f \mathrm{d}t, \\
&= \int_{t_1}^{t_2} \int_{\mathcal{B}} \rho^s \mathbf{a}^s \cdot \delta \mathbf{x}^s + \rho^f \mathbf{a}^f \cdot \delta \mathbf{x}^f \mathrm{d}v \, \mathrm{d}t,
\end{aligned} \tag{18}$$

where integration-by-parts has been used and the first term in the second of (18) vanishes due to the standard assumption in the application of Hamilton's principle that the variations are fixed at the endpoints of the prescribed time interval $[t_1, t_2]$ and

$$\mathbf{a}^{\alpha} = \frac{\partial^2 \chi^{\alpha}(\mathbf{X}^{\alpha}, t)}{\partial^2 t}, \qquad \alpha = s, f.$$

The total potential energy of the binary mixture is given by

$$V = \int_{\mathcal{B}_0} \rho_0^s e^s \left(\mathbf{F}^s, \bar{\rho}^s\right) \mathrm{d}V_0^s + \int_{\mathcal{B}_0} \rho_0^f e^f \left(\bar{\rho}^f\right) \mathrm{d}V_0^f, \tag{19}$$

where the e^s and e^f are the internal energies per unit mass of the solid and fluid respectively. Notice that the internal energy of the solid is treated as a function of deformation and local solid density. This is because the density of the solid constituent can change independently of the deformation due to the internal presence of the pores containing fluid whose volume can change even if $F_{ij}^s = 0$.

Taking the first variation of (19) gives

$$\begin{aligned}
\delta U &= \int_{\mathcal{B}_0} \rho_0^s \left(\frac{\partial e^s}{\partial \bar{\rho}^s} \delta \bar{\rho}^s + \frac{\partial e^s}{\partial F_{ik}^s} \frac{\partial \delta x_i^s}{\partial X_k^s} \right) \mathrm{d}V_0^s + \int_{\mathcal{B}_0} \rho_0^f \frac{\partial e^f}{\partial \bar{\rho}^f} \delta \bar{\rho}^f \, \mathrm{d}V_0^f, \\
&= \int_{\mathcal{B}} \rho^s \left(\frac{\partial e^s}{\partial \bar{\rho}^s} \delta \bar{\rho}^s + \frac{\partial e^s}{\partial F_{ik}^s} \frac{\partial \delta x_i^s}{\partial x_j} F_{jk}^s \right) + \rho^f \frac{\partial e^f}{\partial \bar{\rho}^f} \delta \bar{\rho}^f \mathrm{d}v, \\
&= \int_{\mathcal{B}} \rho^s \frac{\partial e^s}{\partial \bar{\rho}^s} \delta \bar{\rho}^s + \sigma_{ji}^{\prime s} \frac{\partial \delta x_i^s}{\partial x_j} + \rho^f \frac{\partial e^f}{\partial \bar{\rho}^f} \delta \bar{\rho}^f \mathrm{d}v, \\
&= \int_{\mathcal{B}} \rho^s \frac{\partial e^s}{\partial \bar{\rho}^s} \delta \bar{\rho}^s + \rho^f \frac{\partial e^f}{\partial \bar{\rho}^f} \delta \bar{\rho}^f - \frac{\partial \sigma_{ji}^{\prime s}}{\partial x_j} \delta x_i^s \mathrm{d}v + \int_{\partial B} \sigma_{ji}^{\prime s} \hat{n}_j \delta x_i^s \mathrm{d}s, \\
&= \int_{\mathcal{B}} \rho^s \frac{\partial e^s}{\partial \bar{\rho}^s} \delta \bar{\rho}^s + \rho^f \frac{\partial e^f}{\partial \bar{\rho}^f} \delta \bar{\rho}^f - \left(\nabla_{\mathbf{x}} \cdot (\boldsymbol{\sigma}^{\prime s})^{\intercal} \right) \cdot \delta \mathbf{x}^s \mathrm{d}v + \int_{\partial B} \left((\boldsymbol{\sigma}^{\prime s})^{\intercal} \hat{\mathbf{n}} \right) \cdot \delta \mathbf{x}^s \mathrm{d}s,
\end{aligned} \tag{20}$$

where integration-by-parts has been used along with the notation,

$$\sigma_{ji}^{\prime s} = \frac{1}{J^s} P'^s_{ik} F_{jk}^s,$$

$$P_{ij}^{\prime s} = \rho_0^s \frac{\partial e^s}{\partial F_{ij}^s} \bigg|_{\bar{\rho}^s} . \tag{21}$$

which are similar expressions to the Cauchy stress and nominal stress (i.e. the transpose of the first Piola-Kirchhoff stress) and will be further discussed later. The virtual work for this binary mixture can be written as

$$\begin{aligned}
\delta W = &\int_{\mathcal{B}} (\rho^s \mathbf{g} + \mathbf{b}^s) \cdot \delta \mathbf{x}^s + \left(\rho^f \mathbf{g} + \mathbf{b}^f \right) \cdot \delta \mathbf{x}^f \mathrm{d}v \\
&+ \int_{\partial \mathcal{B}} \mathbf{t}_b^s \cdot \delta \mathbf{x}^s - p_b^f \hat{\mathbf{n}} \cdot \delta \mathbf{x}^f \mathrm{d}s,
\end{aligned} \tag{22}$$

where $\mathbf{b}$ is a body force density per unit current volume that is assumed to be identical for both the solid and fluid, e.g. gravity. $\mathbf{h}^s$ and $\mathbf{h}^f$ are drag terms that identify the interactive body forces per unit current volume that the solid and fluid constituents impose one each other. These forces internally balance,

$$\mathbf{b}^s + \mathbf{b}^f = \mathbf{0}.$$

p_b^f is a scalar force per unit current area that is applied opposite the normal to the surface $\partial \mathcal{B}$ on the fluid, i.e. a boundary pressure (compressing the fluid). $\mathbf{t}_b^s$ is a traction per unit current surface area that is applied to the boundary $\partial \mathcal{B}$ of the solid.

Finally, substitute all the results we have for these varations (9), (16), (18), (20), (22), into extended Hamilton's principle (6), and evoke the arbitrariness of $\delta \mathbf{x}^\alpha$, $\delta \phi^\alpha$, and $\delta \bar{\rho}^\alpha$. We have the following equations describing the mixture's motion:

$$
\begin{aligned}
\rho^s \mathbf{a}^s =& \rho^s \mathbf{g} + \mathbf{b}^s - \nabla_{\mathbf{x}} \lambda^s + p \nabla_{\mathbf{x}} \phi^s + \nabla_{\mathbf{x}} \cdot (\boldsymbol{\sigma}'^s)^\intercal, && \text{(23a)}\\
\rho^f \mathbf{a}^f =& \rho^f \mathbf{g} + \mathbf{b}^f - \nabla_{\mathbf{x}} \lambda^f + p \nabla_{\mathbf{x}} \phi^f, && \text{(23b)}\\
\lambda^s =& \phi^s (\bar{\rho}^s)^2 \frac{\partial e^s}{\partial \bar{\rho}^s}, && \text{(23c)}\\
\lambda^f =& \phi^f (\bar{\rho}^f)^2 \frac{\partial e^f}{\partial \bar{\rho}^f}, && \text{(23d)}\\
\lambda^s =& p \phi^s, && \text{(23e)}\\
\lambda^f =& p \phi^f, && \text{(23f)}
\end{aligned}
$$

with the boundary conditions

$$
\begin{aligned}
p_b^f \hat{\mathbf{n}} =& \lambda^f \hat{\mathbf{n}}, && \text{(24a)}\\
\mathbf{t}_b^s =& \left((\boldsymbol{\sigma}'^s)^\intercal - \lambda^s \right) \hat{\mathbf{n}}. && \text{(24b)}
\end{aligned}
$$

Eliminating λ^s and λ^f from (23) and (24), we have

$$
\begin{aligned}
\rho^s \mathbf{a}^s =& \rho^s \mathbf{g} + \mathbf{b}^s - \nabla_{\mathbf{x}} (p \phi^s) + p \nabla_{\mathbf{x}} \phi^s + \nabla_{\mathbf{x}} \cdot (\boldsymbol{\sigma}'^s)^\intercal,\\
\rho^f \mathbf{a}^f =& \rho^f \mathbf{g} + \mathbf{b}^f - \nabla_{\mathbf{x}} \left(p \phi^f \right) + p \nabla_{\mathbf{x}} \phi^f,\\
p =& (\bar{\rho}^s)^2 \frac{\partial e^s}{\partial \bar{\rho}^s} = (\bar{\rho}^f)^2 \frac{\partial e^f}{\partial \bar{\rho}^f},
\end{aligned}
$$

with boundary conditions

$$
\begin{aligned}
p_b^f \hat{\mathbf{n}} =& p \phi^f \hat{\mathbf{n}},\\
\mathbf{t}_b^s =& \left((\boldsymbol{\sigma}'^s)^\intercal - p \phi^s \right) \hat{\mathbf{n}}.
\end{aligned}
$$

And these can be further simplified to

$$
\begin{aligned}
\rho^s \mathbf{a}^s =& \rho^s \mathbf{g} + \mathbf{b}^s - \phi^s \nabla_{\mathbf{x}} p + \nabla_{\mathbf{x}} \cdot (\boldsymbol{\sigma}'^s)^\intercal, && \text{(27a)}\\
\rho^f \mathbf{a}^f =& \rho^f \mathbf{g} + \mathbf{b}^f - \phi^f \nabla_{\mathbf{x}} p, && \text{(27b)}\\
p =& (\bar{\rho}^s)^2 \frac{\partial e^s}{\partial \bar{\rho}^s} = (\bar{\rho}^f)^2 \frac{\partial e^f}{\partial \bar{\rho}^f}, && \text{(27c)}
\end{aligned}
$$

with boundary conditions

$$
\begin{aligned}
p_b^f \hat{\mathbf{n}} =& p \phi^f \hat{\mathbf{n}}, && \text{(28a)}\\
\mathbf{t}_b^s =& \left((\boldsymbol{\sigma}'^s)^\intercal - p \phi^s \right) \hat{\mathbf{n}}. && \text{(28b)}
\end{aligned}
$$

As a remark, the pressure of the fluid is not introduced during the entire derivation, because we only use the local density of fluid to describe the fluid's internal energy. However, equation (27c) shows that p, which is introduced as a Lagrange multiplier in the extended Hamilton's principle, turns out to be the fluid's pressure $(\bar{\rho}^f)^2 \frac{\partial e^f}{\partial \bar{\rho}^f}$ according to the common thermodynamic definition. This result makes sense because p in the binary mixture is the reaction force describing the interaction between fluid and solid through enforcement of volume fraction constraint. This method provides a natural way of introducing fluid pressure in the momentum balance equations for porous media without predefining the effective stress.

Equations (27) and (28) along with appropriate constitutive models for the internal energies and the drag terms as well as statements of conservation of mass can be used to solve for the current position $\mathbf{x}^s$, $\mathbf{x}^f$ and local densities $\bar{\rho}^s$, $\bar{\rho}^f$, therefore establishing a complete theory for this porous media as a binary mixture.

Adding (27a) and (27b) together gives the total momentum balance of the mixture

$$\begin{aligned}\rho^s\mathbf{a}^s + \rho^f\mathbf{a}^f =& \rho\mathbf{g} - \nabla_{\mathbf{x}} p + \nabla_{\mathbf{x}} \cdot (\boldsymbol{\sigma}'^s)^{\intercal}, \\ =& \rho\mathbf{b} + \nabla_{\mathbf{x}} \cdot \left((\boldsymbol{\sigma}'^s)^{\intercal} - p\mathbf{I} \right),\end{aligned}$$

where

$$\rho = \rho^s + \rho^f.$$

Since the term $(\boldsymbol{\sigma}'^s)^{\intercal} - p\mathbf{I}$ represents the total Cauchy stress of the mixture, then we need a total traction boundary condition. We define the total traction $\mathbf{t}_b$ on the boundary ∂B as

$$\mathbf{t}_b = \mathbf{t}_b^s - p_b^f \hat{\mathbf{n}},$$

keeping in mind that by at their introduction into the virtual work $\mathbf{t}_b^s$ was defined as tension positive and p_b^f was defined as compression positive, (i.e. they act in different directions). Now with this definition, we subtract the boundary condition (28a) from (28b) to get

$$\begin{aligned}\mathbf{t}_b =& \left((\boldsymbol{\sigma}'^s)^{\intercal} - p\phi^s - p\phi^f \right) \hat{\mathbf{n}}, \\ =& \left((\boldsymbol{\sigma}'^s)^{\intercal} - p \left(\phi^s + \phi^f \right) \right) \hat{\mathbf{n}}, \\ =& \left((\boldsymbol{\sigma}'^s)^{\intercal} - p \right) \hat{\mathbf{n}},\end{aligned} \tag{29}$$

where the volume fraction constraint $\phi^s + \phi^f = 1$ is used in the last step leading to (29).

It is common to formulate problems in finite deformation poromechanics on the back of the solid skeleton, therefore pulling back to the reference volume of the solid skeleton we have

$$J^s\rho^s\mathbf{a}^s + J^s\rho^f\mathbf{a}^f = J^s\rho\mathbf{b} + \nabla_{\mathbf{X}^s} \cdot \left(\mathbf{P}'^s - pJ^s \left(\mathbf{F}^s\right)^{-\intercal} \right). \tag{30}$$

and the boundary condition

$$\mathbf{T}_b = \left(\mathbf{P}'^s - pJ^s \left(\mathbf{F}^s\right)^{-\intercal} \right) \hat{\mathbf{N}}, \tag{31}$$

where $\mathbf{T}_b = J^s \left(\mathbf{F}^s\right)^{-1} \mathbf{t}_b$ is the total traction per unit reference area applied to the boundary $\partial\mathcal{B}_0$ of the solid and $\hat{\mathbf{N}}$ is a unit vector normal to $\partial\mathcal{B}_0$. Comparing the right-hand side of equation (30) with a finite deformation momentum balance equation, we notice that the term in parenthesis on the right side of (30) plays the role of a nominal stress (i.e. the transpose of the first Piola-Kirchhoff stress) and $\mathbf{P}'^s$ is a type of *effective stress*. Further discussion about the effective stress and meaning of the term $\mathbf{P}'^s$ will made be made in the following section.

4 Correspondence to Biot's Theory

In what follows it will be convenient to introduce the specific volume $\bar{v}^s = 1/\bar{\rho}^s$ then the internal energy of the solid as used in (19) can be equivalently expressed as

$$e^s = e^s(\mathbf{F}^s, \bar{v}^s),$$

and the total differential of e^s is

$$\mathrm{d}e^s = \frac{\partial e^s}{\partial F_{ij}^s}\mathrm{d}F_{ij}^s + \frac{\partial e^s}{\partial \bar{v}^s}\mathrm{d}\bar{v}^s. \tag{32}$$

Also note that using (27c) and the definition of specific volume, we have

$$p = (\bar{\rho}^s)^2 \frac{\partial e^s}{\partial \bar{\rho}^s} = (\bar{\rho}^s)^2 \frac{\partial e^s}{\partial \bar{v}^s}\frac{\partial \bar{v}^s}{\partial \bar{\rho}^s} = -\frac{\partial e^s}{\partial \bar{v}^s}$$

This identifies p as the conjugate variable to $\bar{v}^s$, making it suitable for a Legendre transformation. To switch from $\bar{v}^s$ to p as the independent variable, introduce the Legendre transform $\tilde{e}^s$

$$\tilde{e}^s(\mathbf{F}^s, p) = e^s(\mathbf{F}^s, \bar{v}^s) + p\bar{v}^s. \tag{33}$$

Now, compute the differential of $\tilde{e}^s$

$$\mathrm{d}\tilde{e}^s = \mathrm{d}e^s + p\mathrm{d}\bar{v}^s + \bar{v}^s\mathrm{d}p,$$

and substitute the expression for $\mathrm{d}e^s$ from (32)

$$\mathrm{d}\tilde{e}^s = \left(\frac{\partial e^s}{\partial F_{ij}^s}\mathrm{d}F_{ij}^s + \frac{\partial e^s}{\partial \bar{v}^s}\mathrm{d}\bar{v}^s\right) + p\mathrm{d}\bar{v}^s + \bar{v}^s\mathrm{d}p.$$

Since $\frac{\partial e^s}{\partial \bar{v}^s} = -p$, the terms involving $\mathrm{d}\bar{v}^s$ cancel, i.e.

$$\mathrm{d}\tilde{e}^s = \frac{\partial e^s}{\partial F_{ij}^s}\mathrm{d}F_{ij}^s + \bar{v}^s\mathrm{d}p.$$

Thus, $\tilde{e}^s$ naturally depends on $\mathbf{F}^s$ and p, with

$$\left.\frac{\partial \tilde{e}^s}{\partial F_{ij}^s}\right|_p = \left.\frac{\partial e^s}{\partial F_{ij}^s}\right|_{\bar{v}^s}, \qquad \left.\frac{\partial \tilde{e}^s}{\partial p}\right|_{\mathbf{F}^s} = \bar{v}^s.$$

Rearranging (33) to express e^s in terms of $\tilde{e}^s$

$$e^s = \tilde{e}^s - p\bar{v}^s.$$

Now, compute the derivative of e^s with respect to F_{ij}^s at constant p

$$\left.\frac{\partial e^s}{\partial F_{ij}^s}\right|_p = \left.\frac{\partial \tilde{e}^s}{\partial F_{ij}^s}\right|_p - p\left.\frac{\partial \bar{v}^s}{\partial F_{ij}^s}\right|_p \tag{34}$$

From the differential of $\tilde{e}^s$ and (21), we know

$$\left.\frac{\partial \tilde{e}^s}{\partial F_{ij}^s}\right|_p = \left.\frac{\partial e^s}{\partial F_{ij}^s}\right|_{\bar{v}^s} = \frac{1}{\rho_0^s}P_{ij}'^s.$$

Substitute this into (34) after multiplying through by ρ_0^s and rearrange to get

$$P_{ij}''^s = P_{ij}'^s - p\frac{1}{v_0^s}\left.\frac{\partial \bar{v}^s}{\partial F_{ij}^s}\right|_p \tag{35}$$

where $v_0^s = 1/\rho_0^s$ and $\mathbf{P}''^s$ is the effective stress while holding pressure p constant,

$$P''^s_{ij} = \rho_0^s \left.\frac{\partial e^s}{\partial F^s_{ij}}\right|_p .$$

Furthermore, we note that given the definition of the specific volume and (5) then $\bar{v}^s = \bar{v}^s(J^s)$, using this while solving (35) for $\mathbf{P}'^s$ gives

$$\begin{aligned} P'^s_{ij} &= P''^s_{ij} + p\frac{1}{v_0^s}\left.\frac{\partial \bar{v}^s}{\partial J^s}\right|_p \frac{\partial J^s}{\partial F^s_{ij}} \\ &= P''^s_{ij} + p\frac{1}{v_0^s}\left.\frac{\partial \bar{v}^s}{\partial J^s}\right|_p J^s \left(F_{ji}\right)^{-1} \end{aligned} \tag{36}$$

Substituting (36) into the momentum equation (30)

$$\begin{aligned} J^s\rho^s\mathbf{a}^s + J^s\rho^f\mathbf{a}^f &= J^s\rho\mathbf{b} + \nabla_{\mathbf{X}^s}\cdot\left(\mathbf{P}''^s - \left(1 - \frac{1}{v_0^s}\left.\frac{\partial \bar{v}^s}{\partial J^s}\right|_p\right) pJ^s\left(\mathbf{F}^s\right)^{-\intercal}\right), \\ &= J^s\rho\mathbf{b} + \nabla_{\mathbf{X}^s}\cdot\left(\mathbf{P}''^s - BpJ^s\left(\mathbf{F}^s\right)^{-\intercal}\right), \end{aligned} \tag{37}$$

and likewise into the boundary condition (31)

$$\mathbf{T}_b^s = \left(\mathbf{P}''^s - BpJ^s\left(\mathbf{F}^s\right)^{-\intercal}\right)\hat{\mathbf{N}},$$

and the push-forward

$$\rho^s\mathbf{a}^s + \rho^f\mathbf{a}^f = \rho\mathbf{b} + \nabla_{\mathbf{x}}\cdot\left(\left(\boldsymbol{\sigma}''^s\right)^\intercal - Bp\mathbf{I}\right), \tag{38}$$

where $\boldsymbol{\sigma}''^s$ is the effective Cauchy stress holding p fixed. The boundary condition is

$$\mathbf{t}_b = \left(\left(\boldsymbol{\sigma}''^s\right)^\intercal - Bp\mathbf{I}\right)\hat{\mathbf{n}},$$

where

$$B = 1 - \frac{1}{v_0^s}\left.\frac{\partial \bar{v}^s}{\partial J^s}\right|_p, \tag{39}$$

is the Biot coefficient under finite deformation conditions. The equilibriated forms (i.e. $\mathbf{a}^\alpha = \mathbf{0}$) of (37) and (38) are identical to other theories of finite deformation poromechanics[2]. $\mathbf{P}''^s$ is also the Biot effective nominal stress as commonly used. $\mathbf{P}''^s$ is a preferential stress measure compared to $\mathbf{P}'^s$ because the former is populated with constitutive parameters taken at constant p – an easier parameter to control via experiment than $\bar{\rho}^s$. Notice that $\mathbf{P}''^s$ appeared naturally as a consequence of the variational principle and the Legendre transformation. It was not necessary to preconceptualize or identify it through a constitutive model.

We can demonstrate that under small deformations B is equivalent to the classical Biot coefficient by considering the intrinsic pressure $\bar{p}^s$ applied to the mineral solid alone (i.e. a grain)

$$\bar{p}^s = -\frac{K}{\phi^s}\varepsilon_{kk} - \frac{p}{\phi^s}(1 - B) + p, \tag{40}$$

[2] c.f. [9] eqs. (12) and (13)

where K is the drained ($p = 0$) bulk modulus of the solid skeleton (the mineral solid and pore space), and ε_{kk} is the volumetric strain of the solid skeleton. As before, p is the pressure applied to the solid from the fluid. Appendix B shows the derivation of (40). For small deformations we have $\bar{v}^s \approx \bar{v}_0^s$, $\phi^s \approx \phi_0^s$, and $\varepsilon_{kk} \approx (J-1)$ where J is the Jacobian determinate of the solid skeleton. Appendix A shows that $J^s = J$, so we have

$$\bar{p}^s = -\frac{K}{\phi_0^s}(J-1) + \frac{p}{\phi_0^s v_0^s} \left.\frac{\partial \bar{v}^s}{\partial J}\right|_p + p, \tag{41}$$

using (41) to establish a relationship $J = J(\bar{p}^s)$ and returning to the definition of the Biot coefficient in (39) we apply the chain-rule and substitute terms

$$\begin{aligned} B &= 1 - \frac{1}{v_0^s} \left.\frac{\partial \bar{v}^s}{\partial J}\right|_p, \\ &\approx 1 - \frac{\phi_0^s}{\bar{v}^s} \left.\frac{\partial \bar{v}^s}{\partial \bar{p}^s}\frac{\partial \bar{p}^s}{\partial J}\right|_p, \\ &\approx 1 - \frac{K}{K^s}, \end{aligned} \qquad \blacksquare$$

where

$$\frac{1}{\bar{v}^s}\frac{\partial \bar{v}^s}{\partial \bar{p}^s} = -\frac{1}{K^s},$$

is the definition of the bulk modulus of the mineral solid K^s, and

$$\begin{aligned} \left.\frac{\partial \bar{p}^s}{\partial J}\right|_p &= -\frac{K}{\phi_0^s} + \frac{p}{\phi_0^s v_0^s} \cancelto{\approx 0}{\frac{\partial^2 \bar{v}^s}{\partial J^2}} \\ &= -\frac{K}{\phi_0^s} \end{aligned} \tag{42}$$

proceeds from computing the total differential of (41)

$$\mathrm{d}\bar{p}^s = -\frac{K}{\phi_0^s}\mathrm{d}J + \frac{p}{\phi_0^s v_0^s}\frac{\partial^2 \bar{v}^s}{\partial J^2}\mathrm{d}J + \frac{\mathrm{d}p}{\phi_0^s v_0^s}\frac{\partial \bar{v}^s}{\partial J} + \frac{p}{\phi_0^s v_0^s}\frac{\partial^2 \bar{v}^s}{\partial J \partial p}\mathrm{d}p + \mathrm{d}p.$$

The last term in the first line leading to (42) is approximately zero under the small deformation assumption. Thus proving that B reduces to the classical Biot coefficient. The equilibriated momentum balance (37) can be combined with a finite deformation description of conservation of mass such as shown in [9], Eq. (29) and the system of equations can be solved for $\mathbf{x}^s$ and p.

5 Conclusion

In this paper, we have developed a novel framework that integrates mixture theory with finite deformation poromechanics to derive a nonlinear Biot coefficient from first principles. This approach circumvents the need for constitutive assumptions typically required in traditional methods, such as those based on entropy inequalities or effective stress conceptualizations. By employing the extended Hamilton's principle and introducing a Legendre transformation, we have demonstrated that the Biot coefficient emerges naturally as a reflection of the intrinsic coupling between solid deformation and pore pressure. The nonlinear form of the Biot coefficient suggests an experiment where the density of the mineral solid is monitored with changes in volume to the overall skeleton. Modern imaging techniques such as microCT could be a useful tool in conducting these experiments.

6 Acknowledgements

The first author with like to thank Prof. Chad Landis for a conversation more than a decade before this publication that suggested the use of the Legendre transformation. This work was funded by the U.S. Department of Energy under awards DE-FE0032349 "Hydrogen Storage in Salt Caverns in the Permian Basin: Seal Integrity Evaluation and Field Test" and DE-AR0001864283 assigned control number 2784-1755 "Cyclic Injection for Commercial Seismic-Safe Geologic H2 Production".

A Demonstrate $J = J^s$

Comparing current and reference volume elements leads to

$$J = \frac{\mathrm{d}v}{\mathrm{d}V_0}, \tag{43a}$$

$$\bar{J} = \frac{\mathrm{d}v^s}{\mathrm{d}V_0^s} = \frac{\bar{v}^s}{\bar{v}_0^s}, \tag{43b}$$

where J is a statement of conservation of mass of the solid skeleton and $\bar{J}$ is a statement of conservation of mass for the mineral solid alone. The current volume fraction of the solid ϕ^s can be expressed as

$$\phi^s = \frac{\mathrm{d}v^s}{\mathrm{d}v} = \frac{\bar{J}\mathrm{d}V_0^s}{J\mathrm{d}V_0}, \tag{44}$$

using (43). In the reference configuration we have

$$\phi_0^s = \frac{\mathrm{d}V_0^s}{\mathrm{d}V_0}, \tag{45}$$

substituing (43b) and (45) into (44) gives

$$\phi^s = \frac{\bar{v}^s \phi_0^s}{\bar{v}_0^s J},$$

or

$$\frac{\bar{v}^s}{\bar{v}_0^s} = J\frac{\phi^s}{\phi_0^s}, \tag{46}$$

Finally, substitute (46) into (5) for the solid species s and simplify

$$\begin{aligned} J^s &= \frac{\phi_0^s \bar{\rho}_0^s}{\phi^s \bar{\rho}^s} = \frac{\phi_0^s \bar{v}^s}{\phi^s \bar{v}_0^s} \\ &= J. \end{aligned} \qquad \blacksquare$$

B Deriving the intrensic pressure $\bar{p}^s$

In mixture theory for a binary mixture of a fluid and a solid, the total stress of the mixture is the sum of the partial stresses, i.e.

$$\sigma_{ij} = \phi^s \bar{\sigma}_{ij}^s + \phi^f \bar{\sigma}_{ij}^f, \tag{47}$$

where $\bar{\boldsymbol{\sigma}}^{\alpha}$ is the *intrensic* stress for the α component. The intrinsic stress is the stress within a component as if it were isolated. It represents the "true" stress acting within the material of that compenent. Rearranging (47) we have

$$\phi^s \bar{\sigma}^s_{ij} = \sigma_{ij} - \phi^f \bar{\sigma}^f_{ij}. \tag{48}$$

The total stress of the mixture is given by the term in parenthesis on the right-hand side of (38), i.e. $\boldsymbol{\sigma} = \boldsymbol{\sigma}''^s - Bp\mathbf{I}$. Substitute the total stress into (48) along with $\phi^f = 1 - \phi^s$

$$\phi^s \bar{\sigma}^s_{ij} = \sigma''^s_{ij} - Bp\delta_{ij} - (1-\phi^s)\bar{\sigma}^f_{ij}. \tag{49}$$

Equation (27c) demonstrates that p is the thermodynamic pressure of the fluid, therefore the intrinsic stress on the fluid is $\bar{\sigma}^f_{ij} = -p\delta_{ij}$, substituing into (49),

$$\begin{aligned} \phi^s \bar{\sigma}^s_{ij} &= \sigma''^s_{ij} - Bp\delta_{ij} + p(1-\phi^s)\delta_{ij}, \\ &= \sigma''^s_{ij} + (1-B)p\delta_{ij} - \phi^s p\delta_{ij}. \end{aligned} \tag{50}$$

Substituting a constitutive model for the isotropic elastic solid skeleton $\sigma''^s_{ij} = \left(K - \frac{2}{3}G\right)\varepsilon_{kk}\delta_{ij} + 2G\varepsilon_{ij}$ into (50) and evaluting the the trace of the tensor equation, i.e. $j = i$

$$\begin{aligned} \phi^s \bar{\sigma}^s_{ii} &= \left(K - \frac{2}{3}G\right)\varepsilon_{kk}\delta_{ii} + 2G\varepsilon_{kk} + (1-B)p\delta_{ii} - \phi^s p\delta_{ii}, \\ &= (3K - 2G)\,\varepsilon_{kk} + 2G\varepsilon_{kk} + 3(1-B)p - 3\phi^s p, \\ &= 3K\varepsilon_{kk} + 3(1-B)p - 3\phi^s p, \end{aligned}$$

where the identity $\delta_{ii} = 3$ is used. Now, divide both sides by $-3\phi^s$

$$\begin{aligned} -\frac{1}{3}\bar{\sigma}^s_{kk} &= -\frac{K}{\phi^s}\varepsilon_{kk} - \frac{p}{\phi^s}(1-B) + p, \\ \bar{p}^s &= -\frac{K}{\phi^s}\varepsilon_{kk} - \frac{p}{\phi^s}(1-B) + p, \end{aligned}$$

where the common definition of the hydrostatic pressure $\bar{p}^s = -\frac{1}{3}\bar{\sigma}^s_{kk}$ is introduced in the final step.

References

[1] Francisco Armero. Formulation and finite element implementation of a multiplicative model for coupled poro-plasticity at finite strains under fully saturated conditions. *Computer Methods in Applied Mechanics and Engineering*, 171(3-4):205–241, 1999.

[2] A Bedford and DS Drumheller. A variational theory of porous media. *International Journal of Solids and Structures*, 15(12):967–980, 1979.

[3] Maurice A Biot. General theory of three-dimensional consolidation. *Journal of applied physics*, 12(2):155–164, 1941.

[4] Ronaldo I Borja. On the mechanical energy and effective stress in saturated and unsaturated porous continua. *International Journal of Solids and Structures*, 43(6):1764–1786, 2006.

[5] Olivier Coussy. *Poromechanics*. John Wiley & Sons, 2004.

[6] Gerhard A Holzapfel. *Nonlinear solid mechanics*, volume 24. Wiley Chichester, 2000.

[7] Amos Nur and J_D Byerlee. An exact effective stress law for elastic deformation of rock with fluids. *Journal of geophysical research*, 76(26):6414–6419, 1971.

[8] AW Skempton. Effective stress in soils, concrete and rocks. *Selected papers on soil mechanics*, 1032(3):4–16.

[9] WaiChing Sun, Jakob T Ostien, and Andrew G Salinger. A stabilized assumed deformation gradient finite element formulation for strongly coupled poromechanical simulations at finite strain. *International Journal for Numerical and Analytical Methods in Geomechanics*, 37(16):2755–2788, 2013.